Smriti Rout
IT (M)
58.

Physics Assignment-II

*. Propagation of EM wave in a conducting medium:

Maxwell's eqns are:
$$\text{div } D = \nabla \cdot D = \rho \quad \text{--- (1)}$$
$$\text{div } B = \nabla \cdot B = 0 \quad \text{--- (2)}$$
$$\text{curl } E = -\frac{\partial B}{\partial t} \quad \text{--- (3)}$$
$$\text{curl } H = J + \frac{\partial D}{\partial t} \quad \text{--- (4)}$$

But we know, $D = \varepsilon E$, $B = \mu H$, $J = \sigma E$ & $\rho = 0$

$\therefore$ Maxwell's eqns become:
$$\text{div } E = 0 \quad \text{--- (4)}$$
$$\text{div } H = 0 \quad \text{--- (5)}$$
$$\text{curl } E = -\mu \frac{\partial H}{\partial t} \quad \text{--- (6)}$$
$$\text{curl } H = \sigma E + \varepsilon \frac{\partial E}{\partial t} \quad \text{--- (7)}$$

Taking curl of (6): $\text{curl curl } E = -\mu \frac{\partial}{\partial t}(\text{curl } H)$

Taking value of curl H from (4) and substituting in (6)
$$\text{curl curl } E = -\mu \frac{\partial}{\partial t}\left(J + \frac{\partial D}{\partial t}\right)$$
$$\text{curl curl } E = -\mu \frac{\partial}{\partial t}\left(\sigma E + \varepsilon \frac{\partial E}{\partial t}\right)$$
$$\text{curl curl } E = -\sigma\mu \frac{\partial E}{\partial t} + \varepsilon\mu \frac{\partial^2 E}{\partial t^2} \quad \text{--- (8)}$$

Now taking the value of curl E from (7) & putting in (8)
$$\text{curl curl } H = -\sigma\mu \frac{\partial H}{\partial t} - \varepsilon\mu \frac{\partial^2 H}{\partial t^2}$$

Using vector identity
$$\text{curl curl } A = \text{grad div } A - \nabla^2 A$$

But $\text{div } E = 0$, $\text{div } H = 0$
$$\therefore \text{curl curl } H = -\nabla^2 H$$
$$\& \text{ curl curl } E = -\nabla^2 E$$

$$\Rightarrow \nabla^2 E - \sigma\mu \frac{\partial E}{\partial t} - \varepsilon\mu \frac{\partial^2 E}{\partial t^2} = 0 \quad \text{—(9)}$$

$$\nabla^2 H - \sigma\mu \frac{\partial H}{\partial t} - \varepsilon\mu \frac{\partial^2 H}{\partial t^2} = 0 \quad \text{—(10)}$$

Considering the transverse field in the conducting medium. Let us assume that the fields vary as $e^{ik\cdot r - i\omega t}$ then; Solutions of eqns (9) & (10):

$$E = E_0 e^{i\vec{k}\cdot\vec{r} - \omega t}$$

$$H = H_0 e^{ik\cdot\vec{r} - \omega t}$$

$\vec{K} = \omega/c$ (in free space)

$$k = \alpha + i\beta \quad \text{—(11)}$$

$$\because -k^2 + i\sigma\mu\omega + \mu\varepsilon\omega^2 = 0 \quad \text{—(12)}$$

Squaring eqn (11).

$$k^2 = \alpha^2 - \beta^2 + 2\alpha\beta i \quad \text{—(13)}$$

Now comparing eqns (12) & (13)

$$\therefore \alpha^2 - \beta^2 = \mu\varepsilon\omega^2$$

$$2\alpha\beta = \mu\omega\sigma$$

$$\alpha = \sqrt{\mu\varepsilon}\cdot\omega\left[\frac{\sqrt{(1+(\sigma/\omega\varepsilon)^2} + 1}{2}\right]^{1/2}$$

$$\beta = \sqrt{\mu\varepsilon}\cdot\omega\left[\frac{\sqrt{1+(\sigma/\omega\varepsilon)^2} - 1}{2}\right]^{1/2}$$

The quantity β is a measurement of attenuation and is known as absorption coefficient

$$\left[V = \frac{\omega}{k}\right]$$

$$\therefore V = \frac{\omega}{\alpha} = \frac{1}{\sqrt{\mu\varepsilon}}\left[\frac{\sqrt{1+(\sigma/\omega\varepsilon)^2} + 1}{2}\right]^{-1/2}$$

Case i) For a poor conductor $\frac{\sigma}{\varepsilon} << 1$

$$\alpha = \sqrt{\mu\varepsilon}.\omega \quad \& \quad \beta = 0$$

$$\therefore \quad k = \alpha + i\beta = \sqrt{\mu\varepsilon}.\omega$$

Case ii) For a good conductor $\sigma/\omega\varepsilon >> 1$

$$\alpha \simeq \beta = \sqrt{\mu\varepsilon}\,\omega\sqrt{\left(\frac{\sigma/\omega\varepsilon}{2}\right)} = \sqrt{\frac{\mu\sigma\omega}{2}}$$

$$\therefore k = \alpha + i\beta = (1+i)\sqrt{\frac{\mu\sigma\omega}{2}}$$

* . Propagation of EM wave in a ~~co~~ non-conducting medium: (or simple dielectric medium)

Maxwell's equations :

$$\nabla . D = \rho \quad \text{--- (1)}$$

$$\nabla . B = 0 \quad \text{--- (2)}$$

$$\text{curl} E = \nabla \times E = -\frac{\partial B}{\partial t} \quad \text{--- (3)}$$

$$\text{curl} H = \nabla \times H = J + \frac{\partial D}{\partial t} \quad \text{--- (4)}$$

In an isotropic dielectric (or non-conducting isotropic medium)

$$D = \varepsilon E, \quad B = \mu H, \quad J = \sigma E \quad \& \quad \rho = 0$$

$\therefore$ Maxwell's eqns become :

$$\nabla . E = 0 \quad \text{--- (5)}$$

$$\nabla . B = 0 \quad \text{--- (6)}$$

$$\nabla \times E = -\mu\frac{\partial H}{\partial t} \quad \text{--- (7)}$$

$$\nabla \times H = \varepsilon\frac{\partial E}{\partial t} \quad \text{--- (8)}$$

Taking curl of (7)

$$\text{curl}\,\text{curl} E = -\mu\frac{\partial}{\partial t}(\text{curl} H)$$

Substituting curl H from (8) in above eqn.

$$\text{curl curl } E = -\mu \frac{\partial}{\partial t}\left(\varepsilon \frac{\partial E}{\partial t}\right)$$

$$\text{curl curl } E = -\mu \varepsilon \frac{\partial^2 E}{\partial t^2} \quad \text{—— (9)}$$

Similarly taking curl of (8) & substitute curl E from (7) we get,

$$\text{curl curl } H = -\mu \varepsilon \frac{\partial^2 H}{\partial t^2} \quad \text{—— (10)}$$

But we know, div E = 0 & div H = 0

∴ eqns (9) & (10) become :

$$\nabla^2 E - \mu \varepsilon \frac{\partial^2 E}{\partial t^2} = 0 \quad \text{—— (11)}$$

$$\nabla^2 H - \mu \varepsilon \frac{\partial^2 H}{\partial t^2} = 0 \quad \text{—— (12)}$$

Also, $v = \frac{1}{\sqrt{\mu \varepsilon}} = \frac{1}{\sqrt{\mu_r \mu_0 \varepsilon_r \varepsilon_0}}$ $\left[\because \frac{\mu}{\mu_r} = \mu_0 \ \& \ \frac{\varepsilon}{\varepsilon_r} = \varepsilon_0\right]$

(v: speed of wave)

[where μ_r = permeability of medium
ε_r = relative permittivity of medium]

As, $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ = speed of EM waves in free space.

$$\therefore v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$$

Replacing $\mu\varepsilon$ by $\frac{1}{v^2}$ in eqns (11) & (12)

$$\nabla^2 E - \frac{1}{v^2} \frac{\partial^2 E}{\partial t^2} = 0 \quad \text{—— (13)}$$

$$\nabla^2 H - \frac{1}{v^2} \frac{\partial^2 H}{\partial t^2} = 0 \quad \text{—— (14)}$$

$$\vec{K} = \hat{k}\hat{n} = \hat{n}\,2\pi/\lambda = \frac{\omega}{v}\hat{n}$$

(where $\vec{K}$ = is wave propagation vector)

Relative Dir's of E & H

$\because$ $\nabla.E = 0$ & $\nabla.H = 0$

$\Rightarrow$ $\vec{K}.\vec{E} = 0$ & $\vec{k}.\vec{H} = 0$

$\Rightarrow$ Both the field vectors E & H are perpendicular to the direction of propagation vector $\vec{k}$.

This implies that EM waves in isotropic dielectric are transverse in nature.

Acc. to eqns 7 & 8

$$\text{curl}\, E = -\mu\frac{\partial H}{\partial t} \quad \& \quad \text{curl} H = \varepsilon\frac{\partial E}{\partial t} \quad \text{--- (15)}$$

$$\& \quad \vec{K}\times\vec{H} = -\varepsilon\omega\vec{E} \quad \text{--- (16)}$$

It is now clear that E & H are mutually perpendicular & also they're perpendicular to the dir^n of propagation vector $\vec{K}$.

Phase of E & H & Wave Impedance

$$\vec{H} = \frac{1}{\mu_0\omega}(\vec{K}\times\vec{E}) = \frac{K}{\omega\mu_0}(\hat{\eta}\times\vec{E}) = \frac{1}{\mu_0 c}E$$

In terms of modulus,

$$\left|\frac{E}{H}\right| = \mu_0 c = \mu_0\frac{1}{\sqrt{\mu_0\varepsilon_0}} = \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$\frac{\text{Volt}/m}{\text{Amp}/m} = \frac{\text{Volt}}{\text{Amp}} = \text{Ohm} \qquad Z = \left|\frac{E}{H}\right|$$

$$Z = \sqrt{\frac{\mu_r\mu_0}{\varepsilon_r\varepsilon_0}} = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,.Z_0$$

where Z_0 = wave impedance of free space.

Scanned by CamScanner

Energy flow :

$$\vec{S} = \vec{E} \times \vec{H}$$

(Poynting vector)

$$\vec{S} = \vec{E} \times \left(\frac{\hat{n} \times \vec{E}}{\mu_0 c} \right)$$

$$\vec{S} = \frac{1}{\mu_0 c} (\vec{E} \times \hat{n} \times \vec{E})$$

$$\vec{S} = \frac{1}{\mu_0 C} \hat{n} E^2 = \frac{E^2 \hat{n}}{Z_0}$$

Average value of $\vec{S}$ over a complete cycle :

$$\langle S \rangle = \frac{1}{Z} \langle E^2 \rangle \hat{n}$$

$$= \frac{1}{Z_0} \langle E^2 e^{[i(\vec{k}\cdot\vec{r} - \omega t)]^2} \rangle \hat{n}$$

$$= \frac{1}{Z_0} E^2 \langle \cos^2(\vec{k}\cdot\vec{r} - \omega t) \rangle \hat{n}$$

$$= \frac{1}{Z_0} \cdot \frac{E_0^2}{2} \hat{n}$$

$$\boxed{\langle S \rangle = \frac{E^2_{rms} \cdot \hat{n}}{Z_0}}$$

Ratio of Electric Energy ~~flow~~ density (u_e) to Magnetic ~~field~~ Energy Density (u_m)

$$\frac{u_e}{u_m} = \frac{\frac{1}{2} \varepsilon E^2}{\frac{1}{2} \mu H^2} = \frac{\varepsilon}{\mu} \cdot \frac{E^2}{H^2} = \frac{\varepsilon}{\mu} Z^2 = \frac{\varepsilon}{\mu} \cdot \frac{\mu}{\varepsilon} = 1$$

$$\text{or } \frac{\varepsilon}{\mu} \cdot \frac{E^2}{H^2} = \frac{\varepsilon_0}{\mu} \left| \frac{E}{H} \right|^2$$